

A conservative receiver and the origin of a correlation scale

A finite harmonic receiver with fixed centre velocity has a vanishing long-window displacement-action observable. Random centre velocity instead adds a ballistic term. The exact normal-mode formula identifies the quantity that must lose uniform control before a large receiver can sustain a positive plateau: its low-frequency displacement weight.

1. Hamiltonian and invariant preparation

Let $x, p \in \mathbb{R}^n$, $n \geq 2$, $M = \text{diag}(m_1, \dots, m_n)$ with $m_i > 0$, and let K be a connected spring-network Laplacian. Thus $K = K^T \geq 0$ and $\ker K = \text{span}\{\mathbf{1}\}$. The Hamiltonian and equations are

$$\mathcal{H} = \frac{1}{2}p^T M^{-1}p + \frac{1}{2}x^T Kx, \quad \dot{x} = M^{-1}p, \quad \dot{p} = -Kx.$$

Energy and total momentum are conserved. These linear equations give global smooth trajectories. Write $D = M^{-1/2}KM^{-1/2}$ and choose an orthonormal eigenbasis $e_0, e_1, \dots, e_{n-1}$, with $e_0 = M^{1/2}\mathbf{1}/\sqrt{M_{\text{tot}}}$ and positive eigenvalues ω_j^2 for $j \geq 1$. Define canonical coordinates $q_j = e_j^T M^{1/2}x$, $P_j = e_j^T M^{-1/2}p$.

Fix centre velocity $V_0 = 0$ and centre position. Assign internal mode energies $E_j \geq 0$, and independent phases ϕ_j uniform on $[0, 2\pi)$. Then

$$q_j(t) = \frac{\sqrt{2E_j}}{\omega_j} \cos(\omega_j t + \phi_j), \quad P_j(t) = -\sqrt{2E_j} \sin(\omega_j t + \phi_j).$$

This is an invariant probability ensemble on the internal phase tori, even when frequencies are commensurate. It is a preparation, not a mixing premise. The total energy is $E = \sum_j E_j$, so $|\dot{x}_i| \leq \sqrt{2E/m_i}$ for every trajectory. Choosing $E < \frac{1}{2}(\min_i m_i)c^2$ puts all particles below c in this frame. These are Newtonian springs with a speed-bounded preparation; propagation of signals and Lorentz covariance are separate model requirements.

2. Exact action observable and its limits (C047)

For tagged particle i , let $b_{ij} = e_{ji}/\sqrt{m_i}$, where e_{ji} denotes component i of vector e_j . Then $v_i(t) = \sum_{j \geq 1} b_{ij}P_j(t)$ has zero mean and covariance

$$C_i(t) = \sum_{j \geq 1} w_{ij} \cos(\omega_j t), \quad w_{ij} = b_{ij}^2 E_j \geq 0.$$

Phase averaging gives this identity: distinct modes have zero cross covariance, and one phase has $\mathbb{E}[P_j(t)P_j(0)] = E_j \cos(\omega_j t)$. Integrating the covariance over the displacement square yields, for $\Delta > 0$,

$$h_i(\Delta) = \frac{m_i}{\Delta} \text{Var}[x_i(t + \Delta) - x_i(t)] = \frac{2m_i}{\Delta} \sum_{j \geq 1} \frac{w_{ij}}{\omega_j^2} [1 - \cos(\omega_j \Delta)].$$

The observable has action units and is independent of preparation time t . Set $B_i = \sum_j w_{ij}/\omega_j^2$, with units of length squared. Then

$$0 \leq h_i(\Delta) \leq \frac{4m_i B_i}{\Delta} \longrightarrow 0 \quad (\Delta \rightarrow \infty).$$

At short windows, $h_i(\Delta) = m_i C_i(0)\Delta + O(\Delta^3)$. The finite cosine correlation has persistent memory; its ordinary improper Green–Kubo integral need not exist. The finite-window formula above supplies the relevant limit directly, without replacing the Hamiltonian dynamics by a reversible dissipative generator.

3. Centre motion and the two-body receiver (C047)

An independent random constant centre velocity with variance σ_0^2 adds $m_i\sigma_0^2\Delta$ to h_i . This follows from $x_i(t+\Delta) - x_i(t) = V_0\Delta + \text{internal displacement}$. The velocity ensemble is stationary; for nonzero centre motion the absolute-position ensemble need not be. A deterministic centre velocity contributes zero to the variance and can be removed by a change of frame.

For two particles of masses m, M_r coupled by a spring $k > 0$, put $\mu = mM_r/(m + M_r)$ and $\omega^2 = k/\mu$. At zero total momentum, $x_1 = R + [M_r/(m + M_r)]r$, and assign relative energy E with uniform phase. The tagged response is

$$h_1(\Delta) = \frac{2mE}{k\Delta} \left(\frac{M_r}{m + M_r} \right)^2 [1 - \cos(\omega\Delta)].$$

The spring is a genuine momentum receiver: $m\ddot{x}_1 = -kr$ and $M_r\ddot{x}_2 = kr$. Its reversible exchange produces the oscillatory response rather than the exponential memory law supplied by A02's refreshed bath. Scaling every E_j by a^2 , $0 < a \leq 1$, scales the entire response by a^2 while preserving the equations, conservation laws and speed margin.

4. What the large-receiver test must change (C047)

For a sequence of receivers and windows $\Delta_N \rightarrow \infty$, the bound $h_{i,N}(\Delta_N) \leq 4m_{i,N}B_{i,N}/\Delta_N$ shows that a positive limiting response requires this upper bound to stay positive. In particular, uniform control of $m_{i,N}B_{i,N}$ forces a zero limit. Soft modes can enlarge $B_{i,N}$, but their weights depend on the measured coordinate and the preparation. This is the next selection test.

The finite model yields no physical-time attraction from the invariant preparation: the observable already has no t dependence. Next specify an increasing spring network, its energy allocation and tagged spectral measure, then compare taking receiver size and observation duration to infinity in opposite orders. Derive any surviving constant from these mechanical inputs. Mesh refinement of the same smooth trajectories is a third, distinct limit.

5. Source input and reproduction

Ford–Kac–Mazur, printed p. 505, (2)–(5), supplies the matrix harmonic propagator. Their canonical preparation excludes zero eigenvalues at that stage; ours fixes the centre and uses compact fixed-energy phase tori. Zwanzig, p. 219, (21)–(24), supplies a complementary reduced-description route: eliminate harmonic bath coordinates to obtain a cosine memory kernel. That friction kernel differs from the tagged velocity covariance calculated here. Its continuum replacement must be propagated through the tracer equation before identifying a displacement coefficient.

B23 records the established ingredients, derived finite-network consequence and bounded reading coverage. Run `python3 scripts/harmonic_receiver_checks.py` for the exact modal, two-body and three-body checks. C047 has coordinator proof review and a sequential Luna-low prior-art audit. Its limits concern the stated observable and preparation class.